

Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

More accurate surrogates expose rougher action surfaces and collapse PPO; hybrid separates smooth rollout from physical censoring.

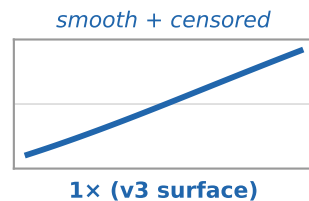
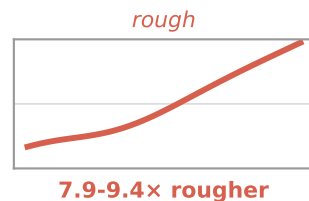
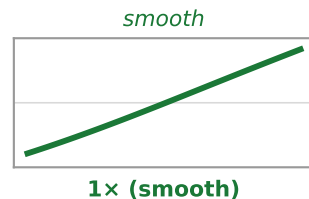
Training backend

Coarse black-box v3
 RMSE_T 1.557 °C

Accurate single-model
v3.5 / matched-v3
 RMSE_T 0.64 / 0.88 °C

Hybrid
v3 rollout + frozen v3.5 censor

Action-response surface



PPO behaviour

modulating
24% saturation

near bang-bang
saturation

modulating
~25% saturation

Live BOPTEST outcome

USABLE

$m_s \approx 0.08$

COLLAPSE

$m_s > 1$

ROBUST

m_s 0.09 pk · 0.04 typ · 85×

RMSE_T = 24 h predictive temperature error · roughness = action-surface curvature ÷ slope (× the usable v3) · m_s = live maintenance score, lower better (pk/typ = peak/typical window; × = throughput vs BOPTEST)